

Disproof of conjecture 00000008434

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Verdict: FALSE

We disprove conjecture 00000008434 as stated. The first clause is refuted by an explicit, genuine block design: AG(3, 2), the unique 3-(8, 4, 1) Steiner quadruple system, has intersection spectrum $\{0, 2\}$, whereas the interval $[s_0, s_1] = [0, 2]$ also contains the value 1, which never occurs. The second clause is refuted by the Pasch configuration, a 1-(6, 3, 2) design with $s_0 = s_1 = 1$ that is not symmetric. Both refutations are finite, elementary, and completely explicit; the first is additionally verified inside Lean 4.

1 The conjecture, quoted verbatim

From `conjectures/00000008434.md`:

English. Definition: The intersection structure of a combinatorial geometry: the intersection spectrum of its blocks. Conjecture: The intersection spectrum is exactly the set of all integer values in the interval $[s_0, s_1]$; and $s_0 = s_1$ if and only if the design is symmetric. (interval characterization of intersection spectra)

The file is bilingual. The Chinese text is reproduced verbatim (in Unicode) in `README.md`; the two versions have identical content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here, and nothing is lost by working from the English below.

2 The statement defines none of its terms

The quoted statement is not a formal mathematical sentence: it introduces names (“combinatorial geometry”, “blocks”, “intersection spectrum”, “symmetric”) without defining any of them, and it never says what s_0 and s_1 are. A disproof therefore has to *pin a reading*, and we record the one adopted here. It is the only standard reading under which the wording has content:

Definition 1 (Pinned reading). A *combinatorial geometry* is a block design: a finite point set P with a family $\mathcal{B}$ of *blocks* (a set system on P). Its *intersection spectrum* is the set of sizes of intersections of *distinct* blocks,

$$S = \{ |B \cap B'| : B, B' \in \mathcal{B}, B \neq B' \},$$

and $s_0 := \min S$, $s_1 := \max S$. A design is *symmetric* when the number of blocks equals the number of points, $b = |\mathcal{B}| = |P| = v$ (equivalently, for 2-designs, $r = k$).

Two points about this reading deserve emphasis.

1. Taking s_0, s_1 to be the *minimum and maximum of the spectrum* is forced by the phrasing: the conjecture asserts an “interval characterization” whose endpoints are s_0, s_1 , and the second clause “ $s_0 = s_1$ iff symmetric” only has content when s_0, s_1 are the two ends of S . (Under this reading that second clause is in fact a *theorem* for genuine 2-designs; see Proposition 10. It is the generality to 1-designs that kills it.)
2. The refutation of clause 1 does not depend on any looseness in the reading. The witness is a full-fledged 3-(8, 4, 1) design — the strongest design hypothesis available in this context — so no “intended” strengthening of “combinatorial geometry” can rescue the claim.

3 The witness: the geometry AG(3, 2)

Definition 2 (The 14 affine planes). Let $P = \mathbb{F}_2^3$, the 8 points. For a nonzero normal vector $a \in \mathbb{F}_2^3 \setminus \{0\}$ and $b \in \{0, 1\}$ put

$$B_{a,b} = \{x \in \mathbb{F}_2^3 : a \cdot x = b\}, \quad a \cdot x = a_1x_1 + a_2x_2 + a_3x_3 \pmod{2}.$$

The point $x = (x_1, x_2, x_3)$ is encoded by the 8-bit mask with bit $x_1 + 2x_2 + 4x_3$ set, so that bitwise AND computes intersections. The family $\mathcal{B} = \{B_{a,b} : a \neq 0, b \in \{0, 1\}\}$ consists of $7 \cdot 2 = 14$ blocks; in sorted mask form they are

$$[15, 51, 60, 85, 90, 102, 105, 150, 153, 165, 170, 195, 204, 240].$$

Proposition 3. $\mathcal{B}$ is a genuine 3-(8, 4, 1) design: $v = 8$, $b = 14$, $k = 4$, every 3-subset of the points lies in exactly one block. Consequently each point lies in $r = 7$ blocks and each pair in $\lambda_2 = (v - 2)/(k - 2) = 3$ blocks, and $\lambda_3 = 1$.

Proof. For fixed $a \neq 0$, the map $x \mapsto a \cdot x$ is a nonzero $\mathbb{F}_2$ -linear functional, so each $B_{a,b}$ is a coset of the kernel $\ker(a \cdot)$, a 2-dimensional subspace; hence $|B_{a,b}| = 2^2 = 4$ and the 14 blocks are affine 2-planes.

Let $T \subset P$ with $|T| = 3$. A line of AG(3, 2) has exactly two points, so no three distinct points are collinear: the three points are affinely independent and span a unique affine 2-subspace, i.e. a unique set of the form $B_{a,b}$. Hence exactly one block contains T , and $\lambda_3 = 1$. The stated λ_2, λ_1 follow from the standard identities

$$\lambda_2 = \frac{v-2}{k-2} \lambda_3 = \frac{6}{2} = 3, \quad \lambda_1 = \frac{v-1}{k-1} \lambda_2 = \frac{7}{3} \cdot 3 = 7.$$

Alternatively, both were checked exhaustively by `reproduce.py` and in Lean (all 56 triples, all 28 pairs, all 8 points). $\square$

This is the unique 3-(8, 4, 1) design (the Steiner quadruple system SQS(8), equivalently the affine geometry AG(3, 2)); the uniqueness is classical and is not needed here.

4 The intersection spectrum of $\text{AG}(3, 2)$

Proposition 4. *For two distinct blocks $B_{a,b} \neq B_{a',b'}$ of Definition 2:*

$$|B_{a,b} \cap B_{a',b'}| = \begin{cases} 0, & a = a' \text{ (the two planes are parallel),} \\ 2, & a \neq a' \text{ (distinct normals).} \end{cases}$$

Hence the spectrum is $S = \{0, 2\}$, with $s_0 = \min S = 0$ and $s_1 = \max S = 2$.

Proof. If $a = a'$ then, since the blocks are distinct, $b \neq b'$; the equations $a \cdot x = b$ and $a \cdot x = b'$ are incompatible, so $B_{a,0} \cap B_{a,1} = \emptyset$. There are exactly 7 nonzero normals a , and each gives exactly one such disjoint pair, so this case occurs 7 times.

If $a \neq a'$, they are linearly independent, so the two affine planes have distinct directions; their intersection is a nonempty affine subspace of dimension $2 + 2 - 3 = 1$, i.e. an affine line, which in $\text{AG}(3, 2)$ has exactly 2 points. This case occurs for all the remaining pairs, of which there are

$$\binom{14}{2} - 7 = 91 - 7 = 84.$$

Thus the spectrum, taken over all 91 unordered pairs of distinct blocks, contains only 0 and 2; both values occur, so $S = \{0, 2\}$. $\square$

The seven disjoint pairs are exactly the seven parallel classes $\{B_{a,0}, B_{a,1}\}$, $a \neq 0$; in mask terms:

$$(15, 240), (51, 204), (60, 195), (85, 170), (90, 165), (102, 153), (105, 150).$$

The 91-pair spectrum table

The next table is the complete intersection-size data. The left block lists, for each candidate intersection size $m \in \{0, 1, 2\}$, the number of unordered pairs $\{B, B'\}$ of distinct blocks with $|B \cap B'| = m$; the right block records whether m lies in the conjectured interval $[s_0, s_1] = [0, 2]$.

intersection size m	number of pairs with $ B \cap B' = m$	$m \in [s_0, s_1] = [0, 2]$?
0	7	yes
1	0	yes
2	84	yes
total	$91 = \binom{14}{2}$	

So exactly $7 + 0 + 84 = 91$ pairs were examined, and the spectrum is $S = \{0, 2\}$: the value 1 is *absent*. The full 14×14 intersection-size matrix (upper triangle; diagonal blank) is displayed below, so that every one of the 91 pairs can be read off directly. Its entries are 2 everywhere off the diagonal except in the seven positions $(1, 14), (2, 13), (3, 12), (4, 11), (5, 10), (6, 9), (7, 8)$, which are 0 — the seven parallel classes.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1		2	2	2	2	2	2	2	2	2	2	2	2	0
2			2	2	2	2	2	2	2	2	2	2	0	2
3				2	2	2	2	2	2	2	2	0	2	2
4					2	2	2	2	2	2	0	2	2	2
5						2	2	2	2	0	2	2	2	2
6							2	2	0	2	2	2	2	2
7								0	2	2	2	2	2	2
8									2	2	2	2	2	2
9										2	2	2	2	2
10											2	2	2	2
11												2	2	2
12													2	2
13														2
14														

5 Clause 1 is false

Theorem 5. *The intersection spectrum S of $\text{AG}(3, 2)$ is not the set of all integer values in the interval $[s_0, s_1]$. Indeed*

$$S = \{0, 2\} \quad \text{but} \quad [s_0, s_1] \cap \mathbb{Z} = \{0, 1, 2\},$$

so the value 1 lies in the conjectured interval and is missing from the spectrum.

Proof. Immediate from Proposition 4: $s_0 = 0$, $s_1 = 2$, so the integer interval is $\{0, 1, 2\}$, while the spectrum omits 1; S is a *strict* subset of the interval, with $|S| = 2 < 3 = |[s_0, s_1] \cap \mathbb{Z}|$. $\square$

Corollary 6. *Conjecture 00000008434, first clause, is false.*

Remark 7. The failure is as small as it can be: the interval has length 2 and exactly one interior value, and that one value is the missing one. The witness is not a boundary case of the definitions — it is the classical 3-(8, 4, 1) design, and the failure comes from clean geometry: two planes in $\text{AG}(3, 2)$ are either parallel (intersecting in 0 points) or meet in an affine line (2 points); there is no way for them to meet in exactly 1 point.

6 Clause 2: false for 1-designs, but a theorem for 2-designs

The second clause of the conjecture is “ $s_0 = s_1$ if and only if the design is symmetric”. It holds for genuine 2-designs, but the conjecture says “design”, and in the general 1-design setting it fails.

Example 8 (Pasch). Let $P = \{1, 2, 3, 4, 5, 6\}$ and

$$\mathcal{B} = \{\{1, 2, 3\}, \{1, 4, 5\}, \{2, 4, 6\}, \{3, 5, 6\}\}.$$

This is the Pasch configuration. It has $b = 4$ blocks of size $k = 3$ on $v = 6$ points; every point lies in exactly $r = 2$ blocks, so it is a 1-(6, 3, 2) design. Every two distinct blocks meet in exactly one point,

$$\{1, 2, 3\} \cap \{1, 4, 5\} = \{1\}, \quad \{1, 2, 3\} \cap \{2, 4, 6\} = \{2\}, \quad \{1, 2, 3\} \cap \{3, 5, 6\} = \{3\},$$

and similarly for the other three pairs; hence $s_0 = s_1 = 1$. But $b = 4 \neq 6 = v$, so the design is *not* symmetric.

Corollary 9. *Clause 2 fails in the generality stated: $s_0 = s_1$ does not imply that the design is symmetric.*

Proposition 10 (The 2-design case, stated to be fair). *Let D be a nontrivial 2 -(v, k, λ) design (so $b \geq v$ by Fisher’s inequality, and the incidence matrix N has rank v). If every two distinct blocks of D meet in the same number μ of points, then D is symmetric, i.e. $b = v$.*

Proof. Let N be the $b \times v$ incidence matrix. Since the off-diagonal entries of $NN^\top$ are the pairwise intersection sizes, the hypothesis says $NN^\top = (k - \mu)I_b + \mu J_b$. This matrix has eigenvalues $k - \mu$ (multiplicity $b - 1$) and $k - \mu + \mu b = k + \mu(b - 1) > 0$. Since $\text{rank}(NN^\top) = \text{rank}(N) = v < b$ when $b > v$, the matrix would have to be singular, forcing $k - \mu = 0$, i.e. $\mu = k$: any two distinct blocks would share all k points of a block, hence be equal, contradicting $b > 1$. Therefore $b > v$ is impossible, and with Fisher’s inequality $b \geq v$ we get $b = v$. $\square$

Thus the second clause is genuinely true for the symmetric-design setting it was presumably intended for; it is the 1-design generality of the wording that makes the “if and only if” false. We report this to delimit the refutation honestly: clause 2 is a defect of the statement’s generality, while clause 1 is refuted even in the strongest, most well-behaved setting.

7 Reproducibility

The complete finite verification is carried out by `reproduce.py` using the Python 3 standard library only:

```
$ python3 reproduce.py
... (builds the 14 planes of  $\mathbb{F}_2^3$ , verifies the 3-(8,4,1)
property, computes all 91 pairwise intersections and their
multiplicities,
checks the Pasch 1-design)
PASS: all checks verified; conjecture 00000008434 is FALSE.
```

It builds the planes from the affine equations (not from the hard-coded mask list), asserts that the generated masks equal the pinned list, verifies all 56 triples have containment count 1, tabulates the 91-pair spectrum, asserts $S = \{0, 2\}$ and that $1 \in [0, 2] \setminus S$, and verifies the Pasch failure. It prints PASS and exits 0 exactly when every check holds (a buggy check was found and fixed this way during preparation).

The LaTeX document is rebuilt with

```
$ tectonic --outdir build main.tex
```

producing `build/main.pdf`.

8 Formalisation

The refutation of clause 1 is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). Blocks are encoded as 8-bit `Nat` masks, the intersection as `Nat.land`, and the spectrum by a structurally recursive function (so it reduces in the kernel). The main statements are

```

theorem blocks_length : blocks.length = 14
theorem triples_length : triples.length = 56
theorem each_triple_in_one_block :
  triples.all (fun T => countIn T == 1) = true
theorem spectrum_length : (spec blocks).length = 91
theorem spectrum_values :
  (spec blocks).all (fun s => s == 0 || s == 2) = true
theorem spec_count_zero : ... filter (s == 0) ... length = 7
theorem spec_count_one : ... filter (s == 1) ... length = 0
theorem spec_count_two : ... filter (s == 2) ... length = 84
theorem s0_eq : s0 = 0
theorem s1_eq : s1 = 2
theorem intervalFull_false : intervalFull = false
theorem intervalSpectrum_false : ¬ IntervalSpectrum
theorem second_clause_fails : ... ∧ ¬ Symmetric pasch.length 6

```

All are proved by kernel `decide` (with `set_option maxRecDepth 1000000`); there is no `sorry`. `lean4/Check.lean` prints `#print axioms` for every theorem; the audit reports only `propext` (and no axioms at all for the pure `Nat` equalities), and in particular no `sorryAx`. See `lean4/README.md` for the statement table and the scope note.

9 Caveats

- **The file defines nothing.** Conjecture 00000008434 is a prose sentence: it fixes no precise meaning for “combinatorial geometry”, “blocks”, “intersection spectrum”, “symmetric”, and it never defines s_0 or s_1 . We pinned the standard reading (Definition 1); the refutation of clause 1 is robust under any reasonable variant because the witness is a genuine 3-design.
- **Clause 2 is only false for 1-designs.** For genuine 2-designs it is a theorem (Proposition 10). We flag this rather than overstate the refutation.
- **Uniqueness of the witness.** $AG(3, 2)$ is the unique 3-(8, 4, 1) design; that classical fact is not required. Everything used is established directly by the finite computations.
- **Lean scope.** The Lean development certifies the finite combinatorial facts (design property, spectrum, absence of 1, negation of clause 1, and the Pasch failure). The general 2-design theorem (Proposition 10) is proved on paper in this document and is not formalised; it is not needed for the disproof.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic --outdir build main.tex`.
- **PDF** — at `build/main.pdf`, produced by the command above.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (name `tlmc8434`, library `Main`, no dependencies), `Main.lean`, `Check.lean`

(#print axioms), and lean4/README.md. lake build exits 0 and the audit reports no sorryAx.